Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

FACILITATOR GUIDE

For

Agentic AI Automation with n8n

TGS Ref No: TGS-2023035977

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 Feb 2023	First version	Tertiary Infotech Academy
2.0	4 September 2024	Added table of content; added Lesson Plan and Instructional Methods in Section 4	Tertiary Infotech Academy
3.0	17 Oct 2025	Update company name	Tertiary Infotech Academy
4.0	12 October 2026	Full revision to the current courseware 'Agentic AI Automation with n8n' (deck v47, LP v3.1, Activities 1–8 and mini capstone) and rebuilt to the WSQ house standard (cover page, version control record, page-numbered table of contents, footer). CP Learning Units, LOs and K&A kept verbatim, with each LU mapped to the n8n topics and activities that deliver it. Lesson plan based on LP v3.1, re-timed to the CP's 22 h training (LU1–LU4 4 h, LU5 6 h) and 2 h assessment (WA-SAQ 4:30–5:30 pm, PP 5:30–6:30 pm on Day 3). Appendix added: v5 assessment questions, suggested answers and marking rubric.	Dr. Alfred Ang
5.0	17 October 2026	Aligned to deck v48, LP v3.2 and LG v6.5: Topic 1 now opens with why agentic AI (history of AI, agentic AI vs AI agents, workflow first vs agent first, AI security) before n8n is introduced; n8n training accounts, sign-in, free Gateway AI credits and n8n Assistant added to Day 1; Activity 4a is now the HR Policy Chatbot built with n8n Assistant; the mini capstone is built with n8n Assistant; slide ranges updated to v48. CP timings unchanged. The assessment appendix moved out of the FG so it can be published: the questions, suggested answers and rubric are now held only in the Assessment Plan Annexes.	Dr. Alfred Ang

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1. Course Overview

Course Title	Agentic AI Automation with n8n
TGS Ref No	TGS-2023035977
Skills Framework	Infocomm Technology (ICT)
TSC Title	Intelligent Reasoning
TSC Code	ICT-ACE-5030-1.1
TSC Category	Development and Implementation
TSC Description	Evaluate, design and build intelligent software systems
TSC Proficiency Level	5
Course Duration	3 days · 24 hours (22 hours training: 11 hrs classroom facilitation + 11 hrs hands-on practical; 2 hours assessment on Day 3)
Daily Timing	9:30 am – 6:30 pm on all three days (1-hour lunch 1:00–2:00 pm not counted; tea breaks within training time)
Mode of Delivery	Classroom (physical), instructor-led: lecture, didactic questioning, demonstration and hands-on n8n practice across the course activities (Activities 1–8) and a mini capstone built with n8n Assistant
Class Size / Ratio	Maximum 20 learners per trainer (CP trainer-to-learner ratio 1:3 min – 1:20 max)
Assessment	Written Assessment – Short-Answer Questions (WA-SAQ) — 60 min, individual, summative, open book Practical Performance (PP) — 60 min, individual, summative, open book

1.1 TSC Knowledge and Abilities

Code	Statement	Assessed by
K1	Cognitive systems	WA-SAQ
K2	Cognitive knowledge representation and techniques	WA-SAQ
K3	Speech comprehension and processing	WA-SAQ
K4	Vision comprehension and processing	WA-SAQ
K5	Natural language comprehension and processing	WA-SAQ
K6	Reasoning systems	WA-SAQ
K7	Search techniques for search-based reasoning applications	WA-SAQ
K8	Optimisation techniques for optimisation reasoning applications	WA-SAQ
K9	Knowledge discovery techniques for reasoning applications	WA-SAQ
K10	Hybrid reasoning systems	WA-SAQ
K11	Data mining framework	WA-SAQ
K12	AI Ethic	WA-SAQ
A1	Identify required cognitive functions based on business needs	PP
A2	Design cognitive applications based on business requirements	PP
A3	Analyse business drivers and application areas of intelligent reasoning systems	PP
A4	Design and apply search techniques to realise expected business outcomes	PP
A5	Decompose complex application scenarios into subproblems	PP
A6	Resolve subproblems by assembling cooperative intelligent subsystems	PP
A7	Design cooperative reasoning modules based on decomposed business outcomes.	PP
A8	Create hybrid reasoning systems by applying suitable techniques and computer programming	PP
A9	Build reasoning systems using hybrid reasoning techniques and sub-modules	PP

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1.2 Abbreviations

Abbreviation	Meaning
LG / FG / AP / LP	Learner Guide / Facilitator Guide / Assessment Plan / Lesson Plan
CP	Course Proposal (TPG-2023044690, approved by SSG)
TSC / K / A	Technical Skills and Competency / Knowledge statement / Ability statement
LU / LO	Learning Unit / Learning Outcome
WA-SAQ / PP	Written Assessment – Short-Answer Questions / Practical Performance
C / NYC	Competent / Not Yet Competent
ASR	Automated Speech Recognition (the CP's subject matter)
LLM / RAG	Large Language Model / Retrieval-Augmented Generation
HITL / MCP	Human-in-the-Loop / Model Context Protocol
TRAQOM	Training Quality and Outcome Measurement survey (SSG)

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2. Learning Units, Learning Outcomes and K&A Mapping

The Learning Units (LUs), Learning Outcomes (LOs) and Knowledge & Abilities (K&A) below are as approved in the Course Proposal; the topic lists reflect the current courseware. Section 2.1 shows where each LU is delivered in the current deck and activities.

Learning Unit	Topics (K&A)	Learning Outcome and K&A Statements
LU1: ASR Systems and Applications in Business	Approved CP content (K1, K2, A1, A2) • ASR systems and their role in businesses (K1, A1) • Audio representation and techniques - spectrogram and Mel Frequency Cepstral Coefficients (MFCC) (K2) • Designing ASR applications based on business requirements (A2) Delivered in courseware v48 as Topic 1: Workflow Automation with n8n (K1, K2, A1, A2) • Why agentic AI: a brief history of AI 2023–2026; generative vs agentic AI; agentic AI vs AI agents • Workflow first vs agent first; AI-assisted workflow creation; where AI agent platforms sit; business value and use cases; five principles of AI security • What n8n is; n8n training accounts, sign-in, the editor, free Gateway AI credits and n8n Assistant • Trigger, data-processing and storage/app action nodes; JSON and expressions; test vs production executions; credentials • Activity 1 — flyer form with QR code (Form → Email); group flyer activity • Activity 2 — capture submissions in a Data Table; Activities 3a/3b — conditional response (Data Table / Google Sheets)	LO1: Identify required ASR functions based on business needs and design ASR applications accordingly K1: Cognitive systems K2: Cognitive knowledge representation and techniques A1: Identify required cognitive functions based on business needs A2: Design cognitive applications based on business requirements
LU2: ASR Fundamentals	Approved CP content (K3, K5, K7, A3, A4) • Introduction to Python and Pytorch packages for speech processing (K3) • Natural language processing with Recurrent Neural Network (RNN), Long Short Term Memory (LSTM) architectures (K5) • Language models and common ASR architectures (A3) • Greedy search and beam search techniques (K7, A4) Delivered in courseware v48 as Topic 2: AI Agents (K3, K5, K7, A3, A4) • Components of an AI agent (model, memory, tools, instructions) • Popular LLM providers; model parameters; memory; tools; good vs bad system instructions • MCP (Model Context Protocol) and MCP in n8n • Activity 4a — HR Policy Chatbot (Ava) on Telegram, built with n8n Assistant from one prompt	LO2: Analyse business drivers to design and apply search techniques in ASR for expected outcomes. K3: Speech comprehension and processing K5: Natural language comprehension and processing K7: Search techniques for search-based reasoning applications A3: Analyse business drivers and application areas of intelligent reasoning systems A4: Design and apply search techniques to realise expected business outcomes

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Learning Unit	Topics (K&A)	Learning Outcome and K&A Statements
	• Activity 4b — give the agent an Employees Data Table tool (HR admin)	
LU3: Siamese Network Systems	Approved CP content (K4, K6, A5, A6) • Object detection with Convolutional Neural Network (CNN) architectures (K4, A5) • Siamese network systems - contrastive loss, triplet loss (K6, A6) Delivered in courseware v48 as Topic 5: Retrieval-Augmented Generation (K4, K6, A5, A6) • Why retrieval rather than a bigger prompt; the RAG pipeline (ingest once, retrieve every question) • Embeddings, vector stores, chunking and top_k; vector databases (in-memory, Pinecone, Supabase, Qdrant) • Activity 7a — RAG chatbot: upload a PDF from a web page, ask in Telegram • Activity 7b — ingest 20 brochures into a vector database; grounded customer-support (CX) agent	LO3: Decompose complex ASR scenarios into subproblems and resolve them using cooperative intelligent subsystems. K4: Vision comprehension and processing K6: Reasoning systems A5: Decompose complex application scenarios into subproblems A6: Resolve subproblems by assembling cooperative intelligent subsystems
LU4: Encoder-Decoder Systems	Approved CP content (K8, K9, A7, A8) • Audio augmentation techniques (K8) • Encoder-Decoder architecture for ASR (A7) • Metrics and Connectionist Temporal Classification (CTC) for ASR evaluation (K9, A8) Delivered in courseware v48 as Topics 3–4: Webhooks; APIs and the HTTP Request Node (K8, K9, A7, A8) • Webhook nodes, webhook vs HTTP Request, GET vs POST, test vs production URL, CORS and Respond to Webhook • Activity 5 — website chatbot via webhook (Investment Advisor) • APIs, anatomy of an HTTP request, status codes, credentials; Twelve Data and NewsAPI • Activity 6 — Finance APIs → Telegram (AI day-trading agent)	LO4: Design cooperative reasoning modules and create hybrid systems in ASR using suitable techniques and programming K8: Optimisation techniques for optimisation reasoning applications K9: Knowledge discovery techniques for reasoning applications A7: Design cooperative reasoning modules based on decomposed business outcomes. A8: Create hybrid reasoning systems by applying suitable techniques and computer programming
LU5: Building and ASR system and Ethical Considerations	Approved CP content (K10, K11, K12, A9) • Speech enhancement techniques (K11) • Building an ASR system (K10, A9) • AI Ethics considerations (K12) Delivered in courseware v48 as Topics 6–7: Security, Human-in-the-Loop & Guardrails; Mini Capstone Project (K10, K11, K12, A9) • Human in, on and out of the loop; approval gates with Send and Wait for Response • Input (pre) and output (post) guardrails around the agent; safe canned replies • Activities 8a–8c — HR Service Portal: leave approval (HITL), dashboard data (GET webhook), AI chatbot with guardrails • Mini capstone — describe an end-to-end business process to n8n Assistant (e.g. a video creation workflow: topic, research, script, production, publish), then review, test, add a control and present	LO5: Build hybrid reasoning systems for ASR using appropriate techniques and sub-modules. K10: Hybrid reasoning systems K11: Data mining framework K12: AI Ethic A9: Build reasoning systems using hybrid reasoning techniques and sub-modules

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2.1 Delivery Map — Learning Units to the Current Courseware

Learning Unit	Delivered in (current deck / activities)	K&A
LU1: ASR Systems and Applications in Business	Day 1 Topic 1 — Workflow Automation with n8n (slides 17–48): why agentic AI and AI security (slides 17–27), what n8n is, training accounts, sign-in, editor, free AI credits and n8n Assistant (slides 28–37), n8n fundamentals (slides 38–48); Activity 1 Flyer Form → Email (slides 49–52), Activity 2 Data Table (slide 53), Activities 3a/3b Conditional Response (slides 55–56). Assessed by PP Task 1 (LO1).	K1, K2, A1, A2
LU2: ASR Fundamentals	Day 1 Topic 2 — AI Agents (slides 57–66); Activity 4a HR Policy Chatbot on Telegram built with n8n Assistant (slides 67–73) and Activity 4b Agent + Data Table Tool (slide 74); Day 1 recap (slide 75). Assessed by PP Task 2 (LO2).	K3, K5, K7, A3, A4
LU3: Siamese Network Systems	Day 2 Topic 5 — Retrieval-Augmented Generation (slides 97–102); Activity 7a RAG Chatbot (slides 103–104) and Activity 7b Vector DB ingestion + CX Agent (slides 105–108). Assessed by PP Task 3 (LO3).	K4, K6, A5, A6
LU4: Encoder-Decoder Systems	Day 2 Topic 3 — Webhooks (slides 76–82) with Activity 5 Website Chatbot via Webhook (slides 83–84); Topic 4 — APIs and the HTTP Request Node (slides 86–92) with Activity 6 Finance APIs → Telegram (slides 93–94); Day 2 recap (slide 95). Assessed by PP Task 4 (LO4).	K8, K9, A7, A8
LU5: Building and ASR system and Ethical Considerations	Day 3 Topic 6 — Security, Human-in-the-Loop & Guardrails (slides 110–113) with Activities 8a–8c HR Service Portal (slides 114–117); Topic 7 — Mini Capstone Project built with n8n Assistant (slides 119–124). Assessed by PP Task 5 (LO5).	K10, K11, K12, A9

3. Facilitator Preparation

3.1 Resources Required

- Trainer slide deck v48 (PPT, 129 slides) and learner slides (PDF), Learner Guide v6.5 and Lesson Plan v3.2 — download from the LMS at https://lms-tms.tertiaryinfotech.com/
- Course activity folders (Activities 1–8, mini capstone, n8n installation with docker-compose.yml) and sample flows: https://github.com/tertiarycourses/TGS-2023035977-Agentic-AI-Automation-with-n8n
- One assigned n8n Cloud training account per learner (password given in class), plus a local Docker Compose installation for use after the course
- Free n8n Gateway AI credits on each training account (OpenAI / Google Gemini API keys optional); n8n Assistant for AI-assisted workflow creation
- Telegram bot accounts (BotFather), Gmail/Outlook and Google Sheets training accounts, Twelve Data and NewsAPI free keys, vector database accounts (Pinecone / Supabase / Qdrant)
- QR code generator (https://alfredang.github.io/qrcodegenerator/) and the Activity 5 / 6 / 7b / 8 reference sites
- Learner laptops (Windows or Mac); 75-inch TV; whiteboard and markers; 1 Gbps Wi-Fi
- TRAQOM digital attendance QR codes (AM, PM and Assessment); WA-SAQ and PP papers; Assessment Summary Record and Appeal Form

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3.2 Before the Class

- Download the current deck (v48), Learner Guide (v6.5), Lesson Plan (v3.2) and the activity folders; confirm the slide ranges against this guide.
- Confirm the n8n training accounts are active (14-day trial running through the class) and the shared password works; update the account table on slide 32 if the accounts changed.
- Check the Gateway credit balance on a training account and test an AI node with Use Gateway credits; keep OpenAI / Gemini keys ready as a fallback.
- Prepare Telegram bot tokens and Google training accounts; plan how any keys are handed out and revoked after class (never in screenshots or shared flows).
- Import and run each activity workflow once end to end (Activities 1–8) so environment issues can be debugged quickly in class.
- Prepare the TRAQOM QR codes for the AM, PM and Assessment digital attendance.
- Pre-load the WA-SAQ and PP papers on the LMS; keep the suggested answers and rubric (Assessment Plan Annexes) with the assessor only.

3.3 Facilitator Roles and Responsibilities

- Deliver every topic and activity in the sequence of the Lesson Plan, using the slide ranges shown.
- Take the mandatory AM, PM and Assessment digital attendance (TRAQOM QR) at the points shown in the lesson plan.
- Check understanding through questioning; coach each learner during the hands-on activities.
- Conduct the Briefing for Assessment, then administer the assessment exactly as specified in the Assessment Plan.
- Record outcomes in the Assessment Summary Record, give feedback, and upload the assessment evidence to the LMS.
- Remind learners to complete the TRAQOM survey and course feedback before they leave.

4. Instructional Methods

Instructional Method	How it is used in this course	Trainer : Learner
Lecture	Concept briefings for each topic using the slide deck — why agentic AI and workflow first, n8n nodes and data flow, AI agent components, webhooks and APIs, RAG, guardrails.	1:3 (min) / 1:20 (max)
Didactic Questioning	Pre-planned questions after each concept segment to check learners can explain why each node, setting or control is needed (e.g. workflow first vs agent first, test vs production URL, embedding dimension, ALLOW/BLOCK guardrail).	1:3 (min) / 1:20 (max)
Demonstration	The trainer builds each activity workflow live in n8n — by hand or with n8n Assistant — before learners attempt it.	1:3 (min) / 1:20 (max)
Practical	Hands-on n8n activities (Activities 1–8, group flyer activity, mini capstone built with n8n Assistant), each debriefed against its expected outcome.	1:3 (min) / 1:20 (max)

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5. Lesson Plan and Facilitator Instructions

Re-timed from the current Lesson Plan (v3.2). LP v3.2 schedules 24 training hours (3 × 8 h) with only a 45-minute 'Assessment and course closing' slot, whereas the CP approves 22 h training (11 h classroom + 11 h practical) and 2 h assessment (WA-SAQ 1 h + PP 1 h), with 4 h each for LU1–LU4 and 6 h for LU5. Each LU receives its CP time (training contact time inclusive of tea breaks and recaps): LU1 9:30–2:30 and LU2 2:30–6:30 on Day 1; LU4 9:30–2:30 and LU3 2:30–6:30 on Day 2; LU5 9:30–4:30 on Day 3, followed by WA-SAQ 4:30–5:30 and PP 5:30–6:30. Day start/end, lunch, topic order, activities and slide ranges follow LP v3.2 / deck v48; activity durations were adjusted to fit.

Day 1 — Why agentic AI and workflow first, workflow automation with n8n, then AI agents

Time	Duration	Topic / Activity and Instructions for Facilitator	Instructional Method	Resources
9:30–9:45	15 min	Welcome, course orientation, trainer and learner introductions, ground rules, learning outcomes; log in to the LMS with OTP and download the course material and GitHub flows; mandatory digital attendance (AM).	Lecture	Slides 1–16, QR
9:45–10:15	30 min	Topic 1 (LU1 · K1, K2, A1, A2) — why agentic AI: a brief history of AI 2023–2026; generative vs agentic AI; agentic AI vs AI agents; workflow first vs agent first; AI-assisted workflow creation; where AI agent platforms sit; business value and use cases; five principles of AI security. Facilitator: ask learners where their own processes sit on the workflow-first / agent-first scale.	Lecture, Didactic Questioning	Slides 17–27
10:15–10:35	20 min	What n8n is; sign in with the assigned n8n training accounts; tour the n8n editor; claim the free Gateway AI credits; n8n Assistant and the social-media campaign demo. Show the local Docker Compose install for after the course.	Demonstration, Practical	Slides 28–37, n8n-installation folder
10:35–10:50	15 min	Key concepts — trigger nodes, data-processing nodes, storage/app actions, JSON and expressions, test vs production executions, credentials. Facilitator: ask learners to name the trigger and the actions in a workflow before building it.	Lecture, Didactic Questioning	Slides 38–48
10:50–11:15	25 min	Activity 1: QR code flyer form (name, email, phone, message) → email to admin via Gmail/Outlook.	Demonstration, Practical	Slides 49–51, Activity 1
11:15–11:50	35 min	Group activity (3–4 pax): design an event flyer (e.g. bowling night), generate a QR code for the form URL and present.	Practical	Slide 52, QR generator
11:50–12:20	30 min	Activity 2: add an n8n Data Table to Activity 1 to capture submissions.	Demonstration, Practical	Slide 53, Activity 2
12:20–1:00	40 min	Activity 3a: conditional response — if attending, save to the Data Table; if not, send a thank-you email (IF node).	Demonstration, Practical	Slide 55, Activity 3
1:00–2:00	60 min	Lunch break	—	Slide 54
2:00–2:30	30 min	Activity 3b: conditional response with Google Sheets / Excel for persistent storage. Topic 1 recap. Facilitator: check each learner's form URL is the production URL and submissions persist.	Practical	Slide 56, Activity 3
2:30–3:10	40 min	Topic 2 (LU2 · K3, K5, K7, A3, A4) — components of an AI agent (LLM, memory, tools, system instruction); LLM providers and model parameters; good vs bad instructions; MCP and MCP in n8n.	Lecture, Didactic Questioning	Slides 57–66
3:10–3:55	45 min	Activity 4a: HR Policy Chatbot (Ava) on Telegram — create the bot with BotFather, build the workflow with n8n Assistant from one prompt, connect the credential, use Gateway credits, publish and chat.	Demonstration, Practical	Slides 67–73, Activity 4

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Time	Duration	Topic / Activity and Instructions for Facilitator	Instructional Method	Resources
		Facilitator: check every learner's agent has a memory node; the reference file is Activity4a-Telegram-HR-Agent.json.		
3:55–4:10	15 min	Tea break	—	—
4:10–6:05	115 min	Activity 4b: give Ava an Employees Data Table tool — reuse Activity 2 data or upload the employee CSV; extend the system instruction and test it; extended practice on memory and tool routing. Facilitator: test that the agent answers only from the tool data and refuses out-of-scope questions (e.g. salaries).	Demonstration, Practical	Slide 74, Activity 4
6:05–6:30	25 min	Day 1 recap, Q&A and mandatory digital attendance (PM).	Didactic Questioning	Slide 75, QR

Day 2 — Webhooks, APIs and HTTP requests, then Retrieval-Augmented Generation

Time	Duration	Topic / Activity and Instructions for Facilitator	Instructional Method	Resources
9:30–9:45	15 min	Day 1 recap and Day 2 objectives; mandatory digital attendance (AM).	Didactic Questioning	QR
9:45–10:15	30 min	Topic 3 (LU4 · K8, K9, A7, A8) — webhooks: inbound URL, webhook vs HTTP Request, GET vs POST, test vs production URL, configuring the Webhook node (CORS), Respond to Webhook.	Lecture, Didactic Questioning	Slides 76–82
10:15–11:20	65 min	Activity 5: website chatbot via webhook — Investment Advisor reference site (Webhook trigger, CORS, AI Agent, Respond to Webhook).	Demonstration, Practical	Slides 83–84, Activity 5
11:20–11:35	15 min	Tea break	—	Slide 85
11:35–11:50	15 min	Selected learners present their live website and chatbot.	Practical	—
11:50–12:15	25 min	Topic 4 (LU4) — APIs and HTTP requests: request anatomy (method, URL, headers, query parameters), status codes, credentials.	Lecture, Didactic Questioning	Slides 86–90
12:15–1:00	45 min	Activity 6 setup and build: sign up for Twelve Data and NewsAPI, store the keys as credentials, import and configure the Finance API workflow.	Demonstration, Practical	Slides 91–93, Activity 6
1:00–2:00	60 min	Lunch break	—	Slide 96
2:00–2:30	30 min	Activity 6 (continued): Finance API → Telegram AI day-trading agent (candles + news, AI Agent, Telegram reply) and the stock dashboard. Topics 3–4 recap. Facilitator: check API keys are in credentials, not hard-coded in the node.	Practical	Slides 94–95, Activity 6
2:30–3:15	45 min	Topic 5 (LU3 · K4, K6, A5, A6) — Retrieval-Augmented Generation: why retrieval, the RAG pipeline, embeddings, vector stores, chunking and top_k, vector databases.	Lecture, Didactic Questioning	Slides 97–101
3:15–3:45	30 min	RAG ingestion walkthrough — upload → embed → vector store → retrieve; set up embedding credentials; the embedding dimension must match the store.	Demonstration	Slides 99–102
3:45–4:00	15 min	Tea break	—	—
4:00–5:00	60 min	Activity 7a: RAG chatbot — upload the IT-Support FAQ PDF from a web page into a vector store, then ask the Telegram bot (knowledge_base tool).	Demonstration, Practical	Slides 103–104, Activity 7
5:00–6:05	65 min	Activity 7b: customer-support RAG agent — ingest 20 course brochures into a vector database (Supabase pgvector, Pinecone or Qdrant) and answer via the Cook & Bake Academy chat widget. Facilitator: ask a question answerable only from the documents and confirm a grounded answer.	Demonstration, Practical	Slides 105–108, Activity 7
6:05–6:30	25 min	Day 2 recap, Q&A and mandatory digital attendance (PM).	Didactic Questioning	Slide 95, QR

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Day 3 — Security, human-in-the-loop and guardrails, mini capstone with n8n Assistant, then assessment

Time	Duration	Topic / Activity and Instructions for Facilitator	Instructional Method	Resources
9:30–9:45	15 min	Day 2 recap and Day 3 objectives; mandatory digital attendance (AM).	Didactic Questioning	QR
9:45–10:25	40 min	Topic 6 (LU5 · K10, K11, K12, A9) — human in, on and out of the loop; how an approval suspends a running flow; the HR Service Portal (three workflows behind one web page).	Lecture, Didactic Questioning	Slides 110–112
10:25–11:25	60 min	Activity 8a: leave application with human-in-the-loop — webhook receives the request, emails the manager an Approve/Reject button (Send and Wait for Response), emails the employee the outcome.	Demonstration, Practical	Slide 114, Activity 8
11:25–11:40	15 min	Tea break	—	—
11:40–12:00	20 min	Selected learners present their Activity 8a approval flow.	Practical	—
12:00–12:25	25 min	Guardrails — input (pre) and output (post) gates around the agent, blocked responses and canned-reply fallbacks; securing credentials and webhooks.	Lecture, Didactic Questioning	Slide 113
12:25–1:00	35 min	Activities 8b and 8c: HR Dashboard Data GET webhook and the AI chatbot with input and output guardrails; wire all three endpoints into the HR Service Portal. Facilitator: test that 'ignore previous instructions…' is blocked and a normal question passes.	Demonstration, Practical	Slides 115–117, Activity 8
1:00–2:00	60 min	Lunch break	—	Slide 118
2:00–2:15	15 min	Topic 7 (LU5) — mini capstone briefing: let n8n Assistant build it; how to write the prompt (trigger, numbered steps, where the human decides); the video creation example (topic → research → script → production → publish) and more example prompts.	Lecture	Slides 119–124
2:15–3:15	60 min	Mini capstone build in small groups with n8n Assistant (supervised lab time): describe the process, review the plan, connect the credentials, test, add a guardrail or approval step. Facilitator: insist every group reviews each generated node before publishing.	Practical	mini-capstone folder
3:15–3:30	15 min	Tea break	—	—
3:30–4:15	45 min	Presentation of mini capstone projects (the prompt, the workflow and a live run); trainer feedback.	Practical, Didactic Questioning	Slide 125
4:15–4:30	15 min	Course recap, Briefing for Assessment, TRAQOM survey and Assessment digital attendance.	Lecture	Slides 14–16, 126–128, QR
4:30–5:30	60 min	Written Assessment — Short-Answer Questions (WA-SAQ): 12 questions, open book (Assessment Plan Annex A).	Assessment	WA-SAQ paper
5:30–6:30	60 min	Practical Performance (PP): FryTech Appliances scenario, 5 n8n tasks (LO1–LO5), open book; workflow screenshots uploaded to the LMS (Assessment Plan Annex C). PM digital attendance.	Assessment	PP paper, QR

Total: 24 hours over 3 days (22 hrs training + 2 hrs assessment), 9:30 am – 6:30 pm, excluding the 1-hour lunch each day.

5.1 Facilitation Notes

- This course is delivered with the current n8n courseware against the CP's Learning Units; Section 2.1 maps each LU to the topics and activities that deliver it.

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- Every activity has a working reference flow in the activity folders — learners who fall behind import it so they can continue with the next topic.
- When learners build with n8n Assistant, have them review every generated node, connect their own credentials and test before they publish; the Assistant drafts the workflow, the learner owns it.
- Never let API keys, bot tokens, training-account passwords or credentials appear in shared flows, screenshots or the chat; store them as n8n credentials, and do not let learners change the training-account password.
- Trial Data Table data is not persistent — use Google Sheets (Activity 3b) or a vector database (Activity 7b) where data must survive.
- A minimum of 75% attendance is required for assessment and funding eligibility.

6. Assessment Overview

6.1 Structure and Duration of Training and Assessment

Component	Duration
Training contact time (inclusive of tea breaks): classroom facilitation 11 hrs + hands-on practical 11 hrs	22 hrs
Written Assessment – Short-Answer Questions (WA-SAQ)	1 hr
Practical Performance (PP)	1 hr
Total course duration	24 hrs

6.2 Summary of Assessment Requirements

Learning Outcome	Assessment Method(s)
LO1: Identify required ASR functions based on business needs and design ASR applications accordingly	WA-SAQ, PP
LO2: Analyse business drivers to design and apply search techniques in ASR for expected outcomes.	WA-SAQ, PP
LO3: Decompose complex ASR scenarios into subproblems and resolve them using cooperative intelligent subsystems.	WA-SAQ, PP
LO4: Design cooperative reasoning modules and create hybrid systems in ASR using suitable techniques and programming	WA-SAQ, PP
LO5: Build hybrid reasoning systems for ASR using appropriate techniques and sub-modules.	WA-SAQ, PP

6.3 Assessment Methods, Duration and Ratio

Method	K&A covered	Questions / Tasks	Duration	Assessor : Candidate
Written Assessment – Short-Answer Questions (WA-SAQ) Individual, summative, open book	K1, K2, K3, K4, K5, K6, K7, K8, K9, K10, K11, K12	Question 1 — K1 Question 2 — K2 Question 3 — K3 Question 4 — K4 Question 5 — K5 Question 6 — K6 Question 7 — K7 Question 8 — K8 Question 9 — K9 Question 10 — K10	60 min	1:3 (min) / 1:20 (max)

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Method	K&A covered	Questions / Tasks	Duration	Assessor : Candidate
		Question 11 — K11 Question 12 — K12		
Practical Performance (PP) Individual, summative, open book	A1, A2, A3, A4, A5, A6, A7, A8, A9	Task 1 (LO1) — A1, A2 Task 2 (LO2) — A3, A4 Task 3 (LO3) — A5, A6 Task 4 (LO4) — A7, A8 Task 5 (LO5) — A9	60 min	1:3 (min) / 1:20 (max)

6.4 Conducting the Assessment

1. Deliver the Briefing for Assessment: purpose, methods, duration, open-book rules, the C/NYC grading and the appeal process.
2. Take the Assessment digital attendance and verify each candidate's identity (name and last 3 digits + alphabet of NRIC/FIN).
3. Issue the question papers (Assessment Plan Annexes) only at the start of each assessment; keep the suggested answers and rubric with the assessor.
4. Invigilate; candidates work individually. Answer clarification questions about the task, never about the answer.
5. Mark each response against the rubric (Assessment Plan Annex E); use oral clarification (1:1, up to 5 minutes) only to close a minor gap.
6. Record C/NYC per LO in the Assessment Summary Record, give feedback, obtain both signatures, and upload the scripts and records to the LMS.
7. A candidate assessed NYC is offered re-assessment in line with the Assessment Plan; explain the appeal procedure if the candidate disagrees.

7. Reference Documents

Document	Purpose
Trainer Slides (PPT) / Learner Slides (PDF) v48	Teaching material for every session
Lesson Plan (LP) v3.2	Session content and activity sequence
Learner Guide (LG) v6.5	Learner reference notes and activity instructions
Assessment Plan (AP) v7	Assessment methods, conditions, records and appeal; the assessment questions, suggested answers and rubric (Annexes A–E)
WA-SAQ and PP papers + answer keys (v5)	Assessment instruments — reproduced in the Assessment Plan Annexes (trainer and assessor only)
Activity folders (Activities 1–8, mini capstone)	Reference n8n workflows, websites and datasets
Course Proposal (CP) TPG-2023044690 (CP_TIPL_ASR_v3)	Approved LUs, LOs, K&A, durations and ratios

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